What Generalizes in Metacognition? Confidence Level, Informativeness, and a Measurement-Reliability Account of Apparent Dissociations

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1 Abstract

Confidence in one's own judgements correlates across cognitive domains, a pattern usually read as evidence for domain-general metacognitive ability. But "confidence" conflates two separable quantities: its LEVEL — a person's average confidence, or set-point — and its INFORMATIVENESS, the metacognitive sensitivity with which confidence discriminates correct from incorrect responses. Because level is measured far more reliably than sensitivity, a naive comparison of their cross-domain correlations manufactures an apparent level-over-sensitivity advantage even when none exists. Using a preregistered, reliability-corrected estimator validated by simulation before it touched the data, we ask which component is domain-general in a four-domain confidence study [1]. Confidence level generalizes strongly across domains (ρ_level = 0.42). Corrected metacognitive sensitivity also correlates across domains (ρ_sens = 0.31, 95% CI excluding zero) — it is not domain-specific. The level-minus-sensitivity gap that a strong dissociation account predicts is 0.113, with a 95% CI that does not exclude zero; the same analysis without correction would have "confirmed" a dissociation (raw gap 0.20, excluding zero). A non-ipsative multilevel decomposition and an external robustness tier across four Confidence Database studies reproduce the pattern: level travels, while the dissociation does not survive attenuation-aware analysis. Apparent dissociations between confidence components in metacognition are, in substantial part, a measurement-reliability artifact.

2 Introduction

People are, to a degree, aware of when they are right and when they are wrong, and they express that awareness as confidence in their own judgements [2]. A striking regularity in the metacognition literature is that confidence appears to travel across otherwise unrelated tasks: someone who reports high or well-calibrated confidence about a perceptual judgement tends to do so about a memory judgement as well. This cross-domain coherence is widely taken as evidence for a domain-general metacognitive faculty — a single self-monitoring capacity that operates over the outputs of many different cognitive systems. Whether such a general capacity exists, and what exactly it would consist of, matters both for theories of metacognition and for the practical hope that a person's self-assessment in one setting tells us something about their self-assessment in another.

The difficulty is that "confidence" is not one thing. A confidence report carries at least two separable quantities. The first is its LEVEL, or set-point: how high a person runs the dial on average — a response-style or scale-use tendency that is largely independent of whether the confidence is deserved. The second is its INFORMATIVENESS, or metacognitive sensitivity: how well a person's confidence discriminates their own correct from incorrect responses. These can dissociate in principle, and they support very different claims. A shared set-point across domains is a domain-general *bias* — a person who simply tends to feel sure

of themselves everywhere. A shared sensitivity across domains is a domain-general *skill* — a genuine self-monitoring ability that generalizes. Much prior work reports cross-domain confidence correlations without cleanly separating these two, so it is often unclear which of the two — bias or skill — the reported generality actually reflects.

This ambiguity would be a minor definitional matter were it not compounded by a measurement hazard that runs in a specific direction. Confidence level is measured almost perfectly from a modest number of trials, because it is just an average. Metacognitive sensitivity, by contrast, is estimated from the covariation between confidence and accuracy within a person, is bounded by the number of trials available, and is measured poorly at realistic trial counts. A difference between two correlations is biased upward when one of the two variables is measured much more reliably than the other. It follows that a raw, uncorrected comparison of level-generalization against sensitivity-generalization will tend to *manufacture* an apparent advantage of level over sensitivity even when the two are equally domain-general in truth. Any honest test of whether the domain-general part of confidence is its level rather than its informativeness must therefore confront differential reliability head-on, rather than reading a raw gap between two correlations at face value. This measurement hazard is the pivot around which the present paper is built.

We take a deliberately conservative, from-scratch approach. Before computing any cross-domain quantity, we build and validate a two-component measurement layer in simulation, requiring that each component's estimator recovers its known planted value and that the two components are separable — that estimating one does not leak into the other. That gate forces two design choices that are not cosmetic: metacognitive sensitivity must be partialed for first-order performance, and one domain pair alone meets the trial-count requirement for a calibrated, reliability-corrected confirmatory test. Around that single preregistered test we place explicitly weaker tiers — a descriptive survey of the remaining domain pairs with frozen bias allowances, a non-ipsative multilevel decomposition that avoids the mean-subtraction artifact known to plague this literature, and an external robustness tier drawn from an independent multi-study database — each carried with attenuation-aware yardsticks rather than raw comparisons. We keep a firewall between building the measurement layer and testing the thesis, and we fix the reporting wording in advance. We state plainly at the outset that a null or partial dissociation is an acceptable, reportable outcome: the design is powered to detect a substantial level-over-sensitivity gap, not a small one, and we report what the data show either way.

This paper makes four contributions. First, a validated two-component measurement framework for confidence with an explicit recovery-and-separability gate, so that the estimators are known to work before they are trusted. Second, a preregistered, reliability-corrected confirmatory test of which component is domain-general in the four-domain study of A. E. Lund, C. M. C. Corrêa, F. Fardo, S. M. Fleming, and M. G. Allen [1]. Third, a non-ipsative multilevel decomposition that asks which component carries the cross-domain covariance without inducing the ipsative artifact of naive mean subtraction. Fourth, an external robustness tier across four multi-task studies from the Confidence Database [3], each read against a simulated null reference rather than pooled. The thesis these tiers converge on is a measurement-lens one: what looks, in prior-style analyses, like a dissociation in which confidence level generalizes and metacognitive sensitivity does not is, once measurement reliability is handled honestly, largely absent as a *distinguishable* effect — while confidence level itself clearly does generalize across domains.

3 Methods

3.1 Primary dataset and provenance

The primary data are the trial-level records of A. E. Lund, C. M. C. Corrêa, F. Fardo, S. M. Fleming, and M. G. Allen [1], a confirmatory multi-domain metacognition study spanning four domains — episodic memory,

visual perception, and two semantic-knowledge judgements (Calories and GDP). We re-fetched the trial file directly from the authors' public repository at a pinned commit (0c38aba4eb…) and verified it before any use: the MD5 checksum matched the value recorded by the authors (500b2e6e1eb0e47e717feba5b912ad9f), and the structure matched expectation — 193,200 trials from 322 subjects across the four domains, with confidence reported on a 1-7 scale. As an external cross-check (never as a primary number) we reproduced per-subject mean confidence against the authors' shipped MLE fits; agreement was high (within-domain correlations 0.998-0.999 across the four domains), the small residual reflecting their undocumented preprocessing. Median trials per subject are 200 for memory and vision and roughly 100 for the two knowledge domains. Following the from-scratch mandate of this project, all analysis code is original to this repository and no computational result from any prior effort was reused or consulted; every reported number is computed in code from the verified raw data.

3.2 The two components and their measures

We operationalize confidence LEVEL (set-point) as the subject's scale-normalized mean confidence within a domain, and metacognitive SENSITIVITY (informativeness) as AUROC2 — the area under the type-2 ROC [2], [4], a model-light index of how well confidence discriminates a person's own correct from incorrect responses. We treat the signal-detection meta-d′/M-ratio family [5] as secondary rather than the spine of the paper, consistent with the model-light posture of this project. Because AUROC2 is confounded by first-order performance (d′), **every sensitivity claim in this paper is d′-partialed**: sensitivity is the residual of AUROC2 after regressing out same-domain d′. This choice is not cosmetic — it was forced by the Step 0 gate (below), where the raw (un-partialed) latent estimator converted a pure d′ correlation into a spurious sensitivity correlation. We do not use gamma as a sensitivity measure: in recovery simulation it leaked the set-point (a level-to-sensitivity contamination), so it was dropped despite its long use as a metacognitive index [6]. The latent cross-domain correlations are estimated with an exchangeable-parcel confirmatory factor model [7] that splits each subject-domain's trials into parcels, giving a measurement model with a built-in reliability correction rather than a single noisy point measure.

3.3 Simulation-gate program (Step 0)

Before testing the thesis we required a foundation gate: prove the estimators recover known planted values, and prove the two components are separable (estimating one does not leak into the other). The gate was run in rounds and we report its failures honestly. The first round established that level is recovered near-perfectly (recovery r ≈ 0.99, no leakage) but that AUROC2 recovery is reliability-capped at realistic trial counts and that the **raw** level-vs-sensitivity contrast manufactures a gap even under a true null — so the naive contrast is invalid as designed. The second round showed that a d′-partialed exchangeable-parcel latent estimator passes all criteria at the 200/200-trial configuration but that raw-latent variants convert the d′ confound into spurious sensitivity correlation — establishing d′-partialing as mandatory. The final round calibrated the corrected estimator on training seeds and evaluated it blind on fresh seeds: at 200 trials per task in both domains the estimator passed every criterion, so the memory × vision pair (the only pair with 200/200 trials) is the sole **confirmatory** pair; other pairs are descriptive with explicit bias bands. The calibration coefficients were then **frozen** (intercept a = 0.0401, slope b = 0.7492) and never refit. A power note carried forward from the gate: at a true gap of 0.4 and the 200/200 configuration, the corrected test excludes zero in about 60% of simulated replicates — the confirmatory test is powered to detect a substantial gap but not a small one.

3.4 Confirmatory design

The single preregistered confirmatory test is the memory × vision pair, chosen mechanically as the only pair meeting the gate's 200/200-trial requirement, and fixed by git timestamp before the cross-domain quantities

were computed. We estimate the latent cross-domain correlation of level (ρ_level) and of d′-partialed sensitivity (ρ_sens) in the parcel-CFA framework, apply the frozen calibration correction to ρ_sens, and form the primary outcome — the gap ρ_level − ρ_sens — with a delta-method confidence interval; a subject bootstrap (B = 2000, calibration held fixed) is the secondary interval. A firewall held until this step: no cross-domain quantity of any kind was computed while the measurement layer and reliabilities were being built. Wording is fixed in advance: a gap whose interval includes zero is reported as **not excluded**, never as absent.

3.5 Descriptive tier

The other five domain pairs did not meet the 200/200 gate and are therefore **descriptive, non-confirmatory**. For each we report ρ_level and the **uncorrected** ρ_sens (no valid calibration exists at their trial configurations), and we attach a frozen systematic-bias band to every gap — ±0.09 for pairs at 200/100 trials and ±0.10 for the Calories × GDP (100/100) pair — carried directly from the gate's measured bias at those configurations. These bands are systematic-bias allowances, not confidence intervals; a descriptive gap is treated as consistent with zero unless it clears both its sampling CI and its band.

3.6 Non-ipsative multilevel decomposition (Step 4)

The classic trap in this literature is the ipsative artifact: subtracting each person's mean confidence mechanically induces negative between-domain correlations, which can masquerade as a domain-specific bias. We therefore do **not** test the thesis by mean subtraction. Instead we fit a trial-level multilevel model [8] that simultaneously estimates a person-general set-point, a person×domain set-point residual, a person-general confidence-resolution slope (Δ confidence for correct vs. incorrect responses, domain-accuracy centered), and a person×domain resolution residual. From the variance components we form two generalizability indices — ICC_level and ICC_sens, the person-general share of set-point and of resolution variance — with cluster-bootstrap intervals (B = 500). This is a variance-share estimand, distinct from the Step 3 latent correlation gap and not a re-test of it. Robustness rungs re-fit with within-domain z-scored confidence (R1) and an ordinal cumulative-link mixed model (R2). As a methods warning we also compute what the discouraged naive mean-subtraction analysis would have produced, on the same trials and in matched simulation, to exhibit the artifact the model is built to avoid. The resolution slope is not d′-partialed; the residual d′ confound is quantified in diagnostics rather than removed, and is flagged wherever the slope is discussed.

3.7 Confidence Database: study selection

For an external robustness tier we used the Confidence Database [3], fetched from OSF and verified file-by-file (255,175,101 bytes across all files; 0 MD5 mismatches). Provenance is drawn from the committed manifest and provenance JSON, not from any prior narrative: the live OSF node has no registered snapshot, so it is pinned by access date and per-file version, and 8 files carry OSF versions above 1 (all four selected studies' data CSVs are version 1). Study selection was mechanical and pre-specified: of 180 parsed study files, 21 have a within-subject multi-task structure, and a four-criterion funnel (a task pair with enough qualifying subjects and trials; a usable confidence scale; both accuracy classes present in every task; no structural disqualifier) selects 4 studies: Arbuzova_2021_expt2 [9], Arbuzova_unpub_3, Bang_2019_Exp2 [10], and Haddara_2022_Expt1 [11]. A firewall again held: no cross-task quantity was computed during selection or measurement.

3.8 Confidence Database: contrast and yardsticks

Taxonomy (binding). This tier tests TASK-generalization and contains exactly one external cross-**domain** pair. Only Arbuzova_unpub_3 memory × visual is cross-domain. Arbuzova_2021_expt2 is a visuomotor condition × condition pair; Bang_2019_Exp2 (coarse vs. fine Gabor) and Haddara_2022_Expt1 (letter

vs. color grids) are cross-**task** pairs within vision. These four studies are never presented as four cross-domain replications.

For each study we correlate confidence level across the task pair and d′-partialed AUROC2 across the same pair (subjects with ≥ 60 clean trials in both tasks, listwise complete; subject bootstrap B = 2000; residualization recomputed inside every resample). Because no calibration correction exists for these configurations, nothing in this tier is confirmatory. Two attenuation-aware yardsticks replace the missing correction: (i) a disattenuation **range** for each correlation (r divided by the geometric mean of the two reliabilities, evaluated at the reliability point estimate and both CI bounds — a sensitivity range, never a corrected point value); and (ii) a per-study **null-reference interval** — the 2.5–97.5% span of the raw gap in a simulated true-null world (true ρ_level = true ρ_sens) matched to each study's n, trials, confidence scale, and reliabilities, quantifying how much raw-gap inflation differential attenuation alone produces. The preregistered reading rule: the Lund pattern **holds** in a study only if r_level exceeds the top of r_sens's disattenuation range **and** the raw gap exceeds the null-reference interval under both null scenarios; one condition met is **partial**; neither is **fails**. No meta-analytic pooling is performed across the four heterogeneous studies. Bang's measures were recomputed from raw with catch trials dropped; Haddara's between-subject feedback groups were derived from raw for a heterogeneity check only.

4 Results

4.1 The measurement floor

Across all four Lund domains, confidence level is measured far more reliably than metacognitive sensitivity (Table 1). Split-half Spearman–Brown reliability [12], [13] of level ranges 0.95-0.99, while AUROC2 reliability ranges only 0.39-0.84. The gap is even starker in trials-needed terms: reaching a reliability of 0.7 takes on the order of 7–15 trials for level but roughly 106 (memory) to 362 (vision) trials for AUROC2. At the parcel level used by the latent model, AUROC2 reliabilities are lower still (memory 0.55, vision 0.27, Calories 0.31, GDP 0.24). This asymmetry is the engine of every inflated raw gap that follows: a difference between two correlations is biased upward when one variable is measured much more reliably than the other.

Table 1: Split-half (Spearman–Brown) reliability of confidence level vs. metacognitive sensitivity (AUROC2), per Lund domain. Sensitivity is measured much less reliably than level in every domain.

	Domain	Level (mean conf.)	Sensitivity (AUROC2)
memory	Memory	0.97	0.84
vision	Vision	0.99	0.59
Calories	Calories	0.95	0.47
GDP	GDP	0.95	0.39

4.2 Confirmatory test: memory × vision

CONFIRMATORY. In the single preregistered pair, confidence level generalizes strongly across domains (ρ_level = 0.42, 95% CI [0.33, 0.51]). Corrected metacognitive sensitivity also correlates across domains and its interval excludes zero (ρ_sens = 0.31, 95% CI [0.08, 0.54]) — sensitivity is not domain-specific either. The primary outcome, the level-minus-sensitivity gap, is 0.113 with 95% CI [−0.138, 0.364] (n = 317): **the null gap is not excluded**. Given roughly 60% power at a true gap of 0.4, this is a "not excluded," not an "absent." The subject bootstrap agrees (gap CI [−0.123, 0.379]).

The corrected result is what keeps this honest. The uncorrected latent sensitivity correlation is 0.27 (95% CI [0.10, 0.44]), giving an uncorrected gap of 0.15; and the fully naive subject-level analysis — raw mean-confidence correlation 0.40 minus raw AUROC2 correlation 0.20 — yields a raw gap of 0.20 with 95% CI [0.05, 0.35] (n = 320) that **excludes zero**. That naive analysis would have affirmed the thesis; the correction machinery, validated in simulation before it touched these data, is exactly what prevented the claim (Table 5, row `naive_gap_exhibit`).

4.3 Descriptive landscape

DESCRIPTIVE (NON-CONFIRMATORY). Across the five non-confirmatory pairs (Table 2, Figure 1), level generalization is uniform (ρ_level range 0.32-0.47) while the gap is heterogeneous and vision-anchored: positive where vision is involved, near zero or negative for the memory × knowledge pairs. No descriptive gap clears both its sampling CI and its systematic-bias band jointly. The most positive gap, vision × GDP (0.30), has a CI that excludes zero, but its lower bound (0.06) lies inside its ±0.09 band, so no nonzero-gap claim is made for it or any descriptive pair. Read through the measurement lens (Figure 2, exploratory), the level-advantage appears where sensitivity is least reliably measured and reverses where it is best measured (the memory pairs) — one measurement fact, not six psychological ones.

Table 2: Descriptive tier (NON-CONFIRMATORY): five domain pairs. rho_sens is uncorrected; each gap carries a frozen systematic-bias band. No gap clears its CI and band jointly.

Pair	rho_level [95% CI]	rho_sens (uncorr.) [95% CI]	gap [95% CI]	band
memory x Calories	0.38 [0.28, 0.47]	0.40 [0.20, 0.59]	−0.02 [−0.24, 0.20]	+/-0.09
memory x GDP	0.32 [0.22, 0.43]	0.46 [0.24, 0.67]	−0.13 [−0.37, 0.11]	+/-0.09
vision x Calories	0.45 [0.36, 0.54]	0.26 [0.05, 0.47]	0.19 [−0.03, 0.42]	+/-0.09
vision x GDP	0.46 [0.37, 0.55]	0.17 [−0.06, 0.39]	0.30 [0.06, 0.54]	+/-0.09
Calories x GDP	0.47 [0.38, 0.56]	0.24 [−0.02, 0.50]	0.24 [−0.04, 0.51]	+/-0.10

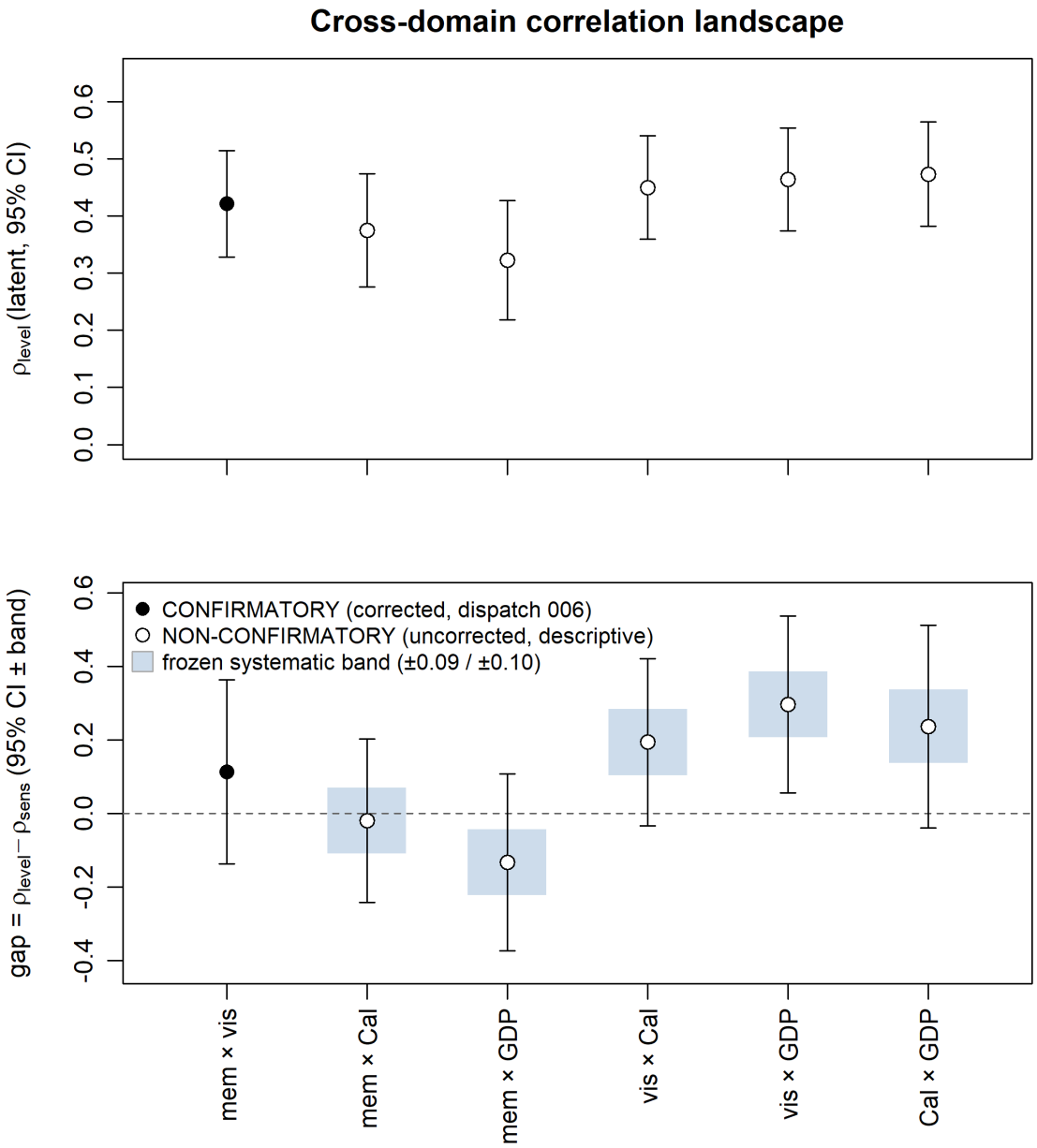


Figure 1: Cross-domain correlation landscape. Top: latent correlation of confidence level (set-point) for each of the six domain pairs, with 95% delta-method CIs. Bottom: the gap (rho_level - rho_sens) per pair. memory x vision (filled point) is the preregistered CONFIRMATORY pair: its rho_sens is calibration-corrected and its gap carries no systematic band. The five open points are NON-CONFIRMATORY / DESCRIPTIVE: their sensitivity correlations are uncorrected (these trial-count configurations failed the blind calibration gate), and each gap therefore carries a shaded rectangle - the frozen systematic-bias allowance (+/-0.09 for 200/100-trial pairs, +/-0.10 for Calories x GDP). The shaded bands are systematic-bias allowances, NOT confidence intervals; whiskers are sampling CIs only. Note vision x GDP: its CI excludes 0, but its lower bound (0.06) lies inside its +/-0.09 band, so no nonzero-gap claim is made for any descriptive pair. The confirmatory result (gap = 0.113, 95% CI [−0.138, 0.364]) does not exclude a null gap.

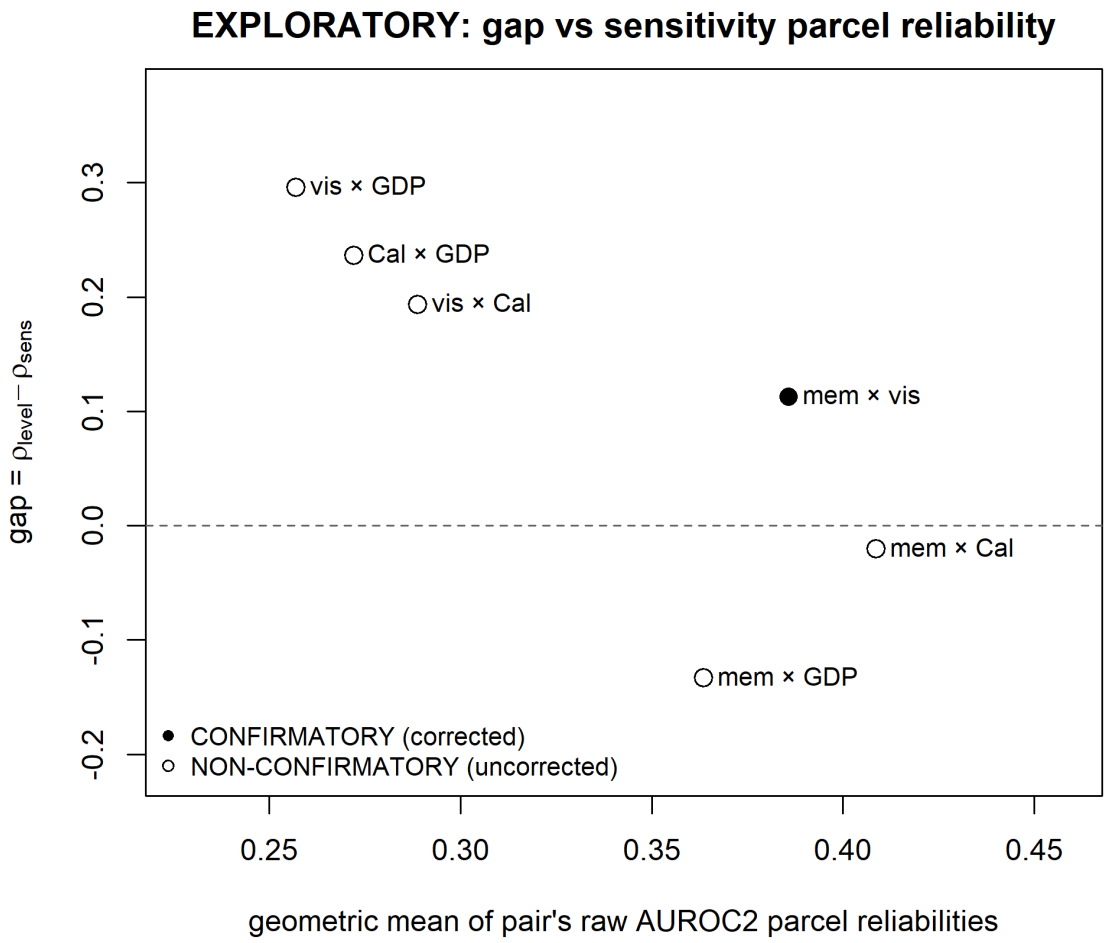


Figure 2: EXPLORATORY: gap vs sensitivity measurement quality. Each pair's gap (as in Figure 1) against the geometric mean of its two domains' raw, pre-partialing AUROC2 parcel reliabilities (memory 0.55, vision 0.27, Calories 0.31, GDP 0.24). No line is fitted, no correlation computed, no test performed: with six non-independent points this display is EXPLORATORY. The measurement-lens reading - the level-advantage appears where sensitivity is least reliably measured and reverses where it is best measured (memory pairs) - is offered as interpretation of residual attenuation, not as a tested result.

4.4 Multilevel decomposition

STEP 4 MULTILEVEL (NON-CONFIRMATORY). The non-ipsative model apportions confidence variance into person-general and person×domain shares. The person-general share is 0.44 for the set-point (ICC_level) and 0.34 for resolution (ICC_sens); their difference is Δ_ICC = 0.11 with 95% cluster-bootstrap CI [0.001, 0.210]. A Δ_ICC of zero sits at the very edge of this interval — we present this as a boundary result and never claim the zero-exclusion; it is directionally consistent with the confirmatory corrected gap (two different estimands, the same "cannot quite exclude zero" verdict). Robustness rungs agree (R1 z-scored: ICC_level 0.44, ICC_sens 0.36; R2 ordinal: 0.43, 0.32). The methods exhibit confirms why mean subtraction was avoided: the naive grand-mean-centered correlations run from −0.67 to 0.08 (mean −0.30) in these data, and in matched simulation the same computation manufactures a mean of about −0.31 under a planted null (mechanical expectation ≈ −1/3), while the multilevel model recovers the planted shares.

4.5 External robustness: the Confidence Database

CONFDB ROBUSTNESS (NON-CONFIRMATORY). This tier tests task-generalization with exactly one external cross-domain pair; the within-vision task pairs and the condition × condition pair are not cross-domain replications. Confidence level travels well: r_level is uniformly positive and precisely estimated across the four studies (0.55–0.87; Table 3, Figure 3). d′-partialed sensitivity is heterogeneous. Because level is measured near-perfectly (split-half 0.97-1.00) and AUROC2 is not (0.62-0.90), a positive raw gap is expected even under a true null, so each raw gap must be read against its simulated null-reference interval (Table 4): at these configurations, differential attenuation plus sampling noise alone can produce raw gaps up to ~0.5 in the small-n studies.

By the preregistered rule, the Lund pattern **holds in 1 of 4** studies, is **partial in 3**, and **fails in 0**. In every study r_level exceeds the top of r_sens's disattenuation range; what separates "holds" from "partial" is whether the raw gap also clears its null reference. Only the single external cross-domain pair does: Arbuzova_unpub_3 memory × visual (r_level = 0.56, 95% CI [0.25, 0.76]; r_sens = −0.10) has a raw gap of 0.551 that exceeds its null-reference upper bounds (0.505 and 0.531) — but only narrowly, by margins of 0.046 and 0.020 against the ρ = 0.30 and ρ = 0.45 references. This single cross-domain pair is therefore consistent with the Lund direction and exceeds its simulated true-null reference, narrowly; a null is not excluded in the three partial studies. Notably, Haddara's within-vision sensitivity correlation is itself high (r_sens = 0.56): metacognitive sensitivity can generalize strongly across near-identical tasks within a domain, which is task generalization, not cross-domain generality.

Table 3: Confidence Database contrast (NON-CONFIRMATORY, robustness). Subject bootstrap B = 2000. Only Arbuzova_unpub_3 memory x visual is cross-domain.

Study	Pair	n	r_level [95% CI]	r_sens (d'-part.) [95% CI]	verdict
Arbuzova_2021_Exp1_2	task1 x task2 (visuomotor, condition x condition)	41	0.55 [0.22, 0.80]	0.24 [−0.08, 0.51]	partial
Arbuzova_unpub_3	memory x visual (CROSS-DOMAIN)	40	0.56 [0.25, 0.76]	−0.10 [−0.35, 0.14]	holds (narrowly)
Bang_2019_Exp2	task1 x task2 (coarse x fine Gabor, within vision)	201	0.57 [0.45, 0.68]	0.07 [−0.05, 0.19]	partial
Haddara_2022_Exp1	task1 x task2 (letter x color grids, within vision)	443	0.87 [0.84, 0.90]	0.56 [0.49, 0.63]	partial

Table 4: Attenuation-aware yardsticks: disattenuation ranges (sensitivity ranges, not corrected point values) and SIMULATION null-reference intervals for the raw gap. Only Arbuzova_unpub_3 exceeds its null-reference interval.

Study	r_level disatt. range	r_sens disatt. range	raw gap [95% CI]	null-ref (rho=.30)	null-ref (rho=.45)
Arbuzova_2021_exp2	[0.52, 0.55]	[0.26, 0.31]	0.14 [−0.22, 0.47]	[−0.36, 0.52]	[−0.31, 0.47]
Arbuzova_unpub_3	[0.57, 0.59]	[−0.25, −0.12]	0.55 [0.19, 0.87]	[−0.32, 0.50]	[−0.24, 0.53]
Bang_2019_Exp2	[0.58, 0.59]	[0.08, 0.09]	0.17 [0.02, 0.32]	[−0.12, 0.25]	[−0.06, 0.26]
Haddara_2022_Exp1	[0.88, 0.88]	[0.64, 0.69]	0.13 [0.09, 0.17]	[−0.08, 0.18]	[−0.04, 0.19]

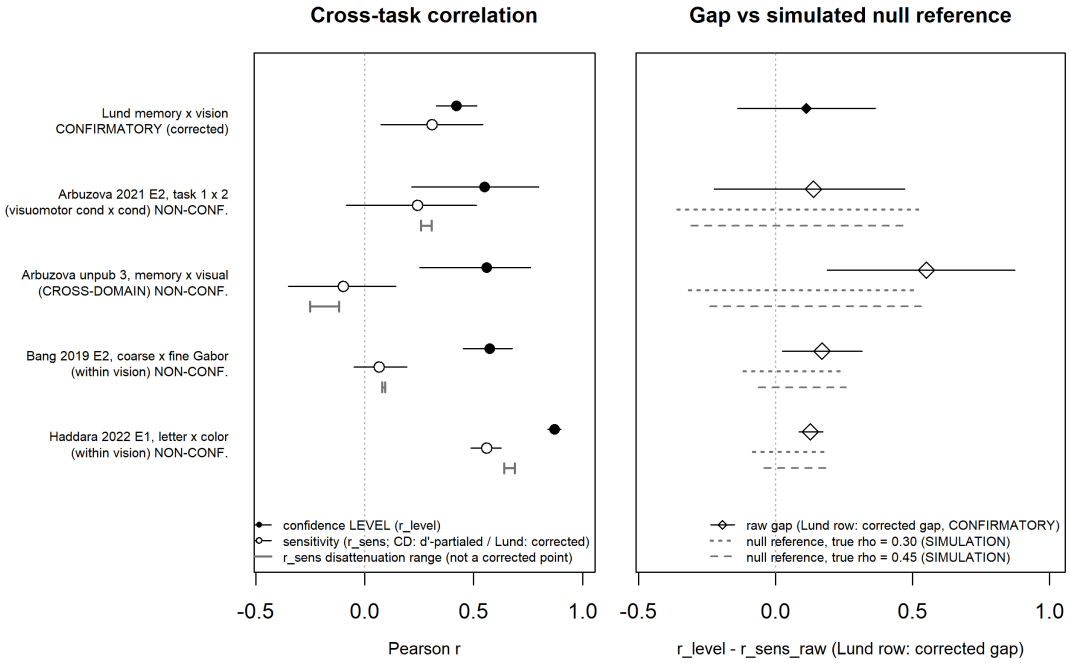


Figure 3: Confidence Database level-vs-sensitivity contrast (NON-CONFIRMATORY, robustness tier) with the Lund confirmatory pair for comparison. Left: cross-task correlations of confidence LEVEL (filled) and metacognitive sensitivity (open; CD rows d'-partialed AUROC2, Lund row calibration-corrected) with 95% CIs; gray brackets are the r_sens disattenuation RANGES - sensitivity ranges, never corrected point values. Right: the RAW gap (r_level - r_sens_raw) per CD study with 95% CI, against SIMULATION null-reference intervals - the 2.5-97.5% span of the raw gap in a true-null world (true r_level = true r_sens = 0.30 dotted / 0.45 dashed) matched to each study's n, trials, scale, and reliabilities. Raw gaps are inflated by differential attenuation (level split-half reliabilities 0.97-1.00 vs AUROC2 0.62-0.90), so a positive raw gap is expected even under a true null; only Arbuzova_unpub_3 memory x visual exceeds its null-reference intervals (raw gap 0.551 vs upper bounds 0.505 / 0.531). This tier tests TASK-generalization with exactly one external cross-domain pair: only Arbuzova_unpub_3 memory x visual is cross-DOMAIN; Arbuzova_2021_expt2 is a visuomotor condition x condition pair, and Bang_2019_Exp2 and Haddara_2022_Expt1 are within-vision task pairs - these four studies are not four cross-domain replications. The Lund row is the preregistered CONFIRMATORY (corrected) result: its gap (0.113, 95% CI [−0.138, 0.364]) does not exclude a null. Null-reference intervals are a simulation reference, not a correction and not a test.

4.6 Claim-status summary

Table 5 collects every quantitative claim with its tier, estimate, and frozen verdict wording (full machine-generated table with sources and CIs: `results/step17_claim_status.md`).

Table 5: Claim-status summary (generated by analysis/17_claim_status.R from committed CSVs). Every value is assert-guarded in code; verdict wording is preregistered.

#	Claim	Tier	Estimate	Verdict (frozen)
1	gap_confirmatory	CONFIRMATORY	0.113 [−0.138, 0.364]	null gap not excluded
2	rho_level_confirmatory	CONFIRMATORY	0.422 [0.328, 0.515]	confidence level is domain-general
3	rho_sens_confirmatory	CONFIRMATORY	0.308 [0.075, 0.542]	corrected sensitivity correlation excludes zero - sensitivity is not domain-specific either
4	naive_gap_exhibit	METHODS EXHIBIT	0.2 [0.051, 0.353]	the uncorrected analysis would have affirmed the thesis; the correction machinery prevented that claim
5	descriptive_landscape	DESCRIPTIVE (NON-CONFIRMATORY)	–	heterogeneous, vision-anchored; no nonzero-gap claim for any descriptive pair
6	step4_decomposition	STEP4 MULTI-LEVEL (NON-CONFIRMATORY)	0.105 [0.001, 0.21]	a Delta_ICC of zero sits at the very edge of the interval; directionally consistent with the confirmatory corrected gap; the zero-exclusion is never claimed
7	ipsative_exhibit	METHODS EXHIBIT	−0.667	grand-mean centering manufactures negative within-subject correlations; asymmetric under unequal trial counts
8	confdb_tier	CONFDB ROBUSTNESS (NON-CONFIRMATORY)	–	holds 1/4, partial 3/4, fails 0/4; the only external cross-domain pair is Arbuzova_unpub_3 memory x visual (consistent with the Lund direction, exceeds its simulated true-null reference narrowly); within-vision task pairs and the condition x condition pair are NOT cross-domain replications
9	reliability_floor	METHODS (NON-CONFIRMATORY)	–	the measurement asymmetry that inflates raw gaps is universal in these data

5 Discussion

Read through the measurement lens, these results tell a two-sided story that resists both of the overclaims available here. On one side, confidence level generalizes strongly and reliably across domains: in the preregistered confirmatory pair its cross-domain correlation is 0.42, and it is uniformly positive everywhere else we look. On the other side, once metacognitive sensitivity is corrected for its poor reliability, it too correlates across domains, with an interval that excludes zero (0.31) — so sensitivity is not domain-specific either. The quantity a strong dissociation account cares about, the level-minus-sensitivity gap, is 0.113 with a 95% CI of [−0.138, 0.364] that does not exclude zero. The contrast that makes the point sharpest is internal to the same data: the naive subject-level analysis yields a gap of 0.20 whose interval *excludes* zero and would have affirmed a dissociation, whereas the reliability-corrected analysis of the identical subjects and domains does not. The apparent dissociation that prior-style analyses would report is, in substantial part, a differential-reliability artifact. This cuts in both directions at once: against overclaiming a level-over-sensitivity dissociation, and against the opposite conclusion that metacognitive sensitivity fails to generalize.

The descriptive landscape across the remaining domain pairs is best read the same way — as one measurement fact wearing several costumes rather than several psychological facts. The level-minus-sensitivity gap is heterogeneous across pairs: it is most positive where vision is involved and reverses to near zero or negative for the pairs that involve memory. That pattern tracks measurement quality rather than psychology. Metacognitive sensitivity is estimated most reliably in memory (parcel reliability 0.55) and least reliably in the other domains (vision 0.27, Calories 0.31, GDP 0.24); the apparent level advantage is largest exactly where sensitivity is measured worst, and it shrinks or reverses where memory's better-measured sensitivity enters the pair. We present this as an attenuation *interpretation*, not a tested result: the gap-versus-reliability display (Figure 2) fits no line and performs no test, and with a handful of non-independent pairs it could not. It is worth being explicit about the one descriptive pair whose sampling interval excludes zero, vision × GDP: its lower bound still lies inside its frozen systematic-bias band, so even there we make no nonzero-gap claim. The heterogeneity that a psychological account would have to explain domain by domain is, on this reading, the shadow of a single asymmetry in how well the two components can be measured.

The external tier from the Confidence Database adds a genuine but bounded piece of evidence, and its structure must be stated carefully. Confidence level again travels well: its cross-task correlation is positive and precisely estimated in every one of the four studies (0.55–0.87). Metacognitive sensitivity is where the taxonomy matters. Only one of the four pairs is genuinely cross-**domain** (Arbuzova_unpub_3, memory × visual); the others are a condition × condition pair and two within-vision task pairs, which test task-generalization, not cross-domain generality, and are never presented as cross-domain replications. By the preregistered rule the Lund pattern holds in that single cross-domain pair only narrowly: its raw gap clears its simulated null reference by margins of just 0.046 and 0.020, so the replication is directional rather than decisive, and a null is not excluded in the three task-generalization pairs. Tellingly, one of those within-vision pairs shows a strong sensitivity correlation (0.56): metacognitive sensitivity clearly *can* generalize across near-identical tasks within a domain. That is a demonstration of task generalization, and it should not be mistaken for evidence of cross-domain metacognitive skill — the two are different claims, and only the former is on display here.

These results also let us reconcile the paper with the older claim that metacognition harbours a domain-general trait. The signal that most robustly survives across domains, in every tier, is the confidence set-point — the level component. The multilevel decomposition tells the same story from the variance-share side: the person-general share of the set-point (0.44) sits above that of confidence resolution (0.34), with their

difference a boundary result that never claims to exclude zero. What this suggests is that a good deal of what has been described as "domain-general metacognition" may be domain-general confidence *bias* — shared scale-use, a stable set-point — rather than a shared metacognitive *ability*. That is a more modest object than a general self-monitoring faculty, and it is the one the present data most clearly support; it does not deny that sensitivity generalizes, only that the evidence here cannot cleanly separate a generalizing sensitivity from a generalizing set-point once reliability is handled honestly.

What would settle the question the confirmatory test leaves open is more measurement, not more theory. The corrected test is powered to detect a substantial gap and not a small one, and metacognitive sensitivity sits on a reliability floor throughout these data. Lifting sensitivity off that floor — more trials per task, so that its reliability approaches the near-ceiling of level — would let a reliability-corrected gap be estimated with far less uncertainty. So would more genuinely cross-domain pairs, rather than the largely within-domain task pairs that the available databases mostly offer, and higher-powered confirmatory designs built for a small effect. The reusable contribution of this paper is not the particular verdict in this dataset but the machinery that produced it: a recovery-and-separability gate, an explicit reliability correction validated before use, and simulated null references in place of raw comparisons.

The moral generalizes beyond these four domains. Wherever two metacognitive quantities are measured with very different reliability — and confidence level and metacognitive sensitivity almost always are — comparing their cross-domain correlations without correction will invite a manufactured dissociation. The honest version of the question is not "does level generalize more than sensitivity?" but "does it generalize more than differential attenuation alone would predict?" In these data, once that question is asked properly, confidence level clearly generalizes, and the apparent gap between the two components does not survive.

6 Limitations

Several limitations bound these conclusions. First, the confirmatory test is powered to detect only a substantial dissociation: at a true gap of 0.4 the corrected test excludes zero roughly 60% of the time, so the null gap we report is "not excluded" rather than evidence of no dissociation — the design cannot distinguish a true null from a small gap it is underpowered to find. Second, metacognitive sensitivity is weakly measured everywhere in these data (split-half AUROC2 reliability far below that of confidence level); we therefore present disattenuation only as a range, never as a corrected point estimate, and treat the reliability asymmetry as a systematic feature rather than a nuisance to be fully removed. Third, the multilevel Δ_ICC is a boundary result: a Δ_ICC of zero sits at the very edge of its interval, within Monte-Carlo error of the bound, so we do not claim a zero-exclusion. Fourth, the confirmatory test rests on a single dataset and a single 200/200-trial domain pair (memory × vision); the other pairs are explicitly descriptive. Fifth, the external cross-domain evidence rests on a single small study (Arbuzova_unpub_3, n = 40) that clears its simulated null reference only narrowly (margins 0.046 and 0.020), and the smaller margin shrinks further under an independently constructed null simulator — so this replication is directional, not decisive; the remaining external pairs are within-vision task pairs or a condition pair and test task-generalization, not cross-domain generality. Sixth, the null-reference simulator matches each study's reliabilities and means but not higher moments, and uses fixed per-task accuracy with equally spaced latent thresholds. Seventh, the Confidence-Database disattenuation applies AUROC2 reliabilities to the d′-partialed residual, whose reliability is not identical to that of raw AUROC2. Eighth, first-order sensitivity (d′) is itself poorly measured outside the memory domain, and the multilevel resolution slope is not d′-partialed (the residual confound is quantified, not removed). Finally, gamma was excluded as a sensitivity measure because it leaked the set-point in recovery simulation, and we treat the meta-d′/M-ratio family as secondary; no meta-analytic pooling is performed across the heterogeneous Confidence-Database studies.

7 Data and code availability

All analysis code is original to this project and public at `https://github.com/dondon20201/paper2`; the analysis runs end-to-end under a renv lockfile (R 4.6.1) [14]. Every reported number is written to a file under `results/` with a name traceable from the text, and each figure's underlying data is saved alongside it (`results/fig_landscape_data.csv`, `results/figS_gap_vs_reliability_data.csv`, `results/fig_confdb_contrast_data.csv`).

Primary data. A. E. Lund, C. M. C. Corrêa, F. Fardo, S. M. Fleming, and M. G. Allen [1], file `data_summary/metacognition_trialData_master.csv` from the repository `embodied-computation-group/dg-metacognition`, pinned at commit 0c38aba4eb… and verified by MD5 (500b2e6e1eb0e47e717feba5b912ad9f). The raw file is not redistributed here (provenance in `data/raw/lund2025/README.md`).

Secondary data. The Confidence Database [3] from OSF node `s46pr`, verified file-by-file against OSF MD5s (255,175,101 bytes, 0 mismatches); the live node has no registered snapshot, so it is pinned by access date and per-file OSF version in `results/step11_CONFDB_manifest.csv`. Raw Confidence Database files (~255 MB) are not redistributed here.

8 References

Bibliography

[1] A. E. Lund, C. M. C. Corrêa, F. Fardo, S. M. Fleming, and M. G. Allen, "Domain generality in metacognitive ability: A confirmatory study across visual perception, episodic memory, and semantic memory," *Journal of Experimental Psychology: Learning, Memory, and Cognition*, vol. 51, no. 10, pp. 1594–1605, 2025, doi: 10.1037/xlm0001462.

[2] S. M. Fleming and H. C. Lau, "How to measure metacognition," *Frontiers in Human Neuroscience*, vol. 8, p. 443, 2014, doi: 10.3389/fnhum.2014.00443.

[3] D. Rahnev, K. Desender, A. L. F. Lee, W. T. Adler, and others, "The Confidence Database," *Nature Human Behaviour*, vol. 4, no. 3, pp. 317–325, 2020, doi: 10.1038/s41562-019-0813-1.

[4] S. J. Galvin, J. V. Podd, V. Drga, and J. Whitmore, "Type 2 tasks in the theory of signal detectability: Discrimination between correct and incorrect decisions," *Psychonomic Bulletin & Review*, vol. 10, no. 4, pp. 843–876, 2003, doi: 10.3758/BF03196546.

[5] B. Maniscalco and H. Lau, "A signal detection theoretic approach for estimating metacognitive sensitivity from confidence ratings," *Consciousness and Cognition*, vol. 21, no. 1, pp. 422–430, 2012, doi: 10.1016/j.concog.2011.09.021.

[6] T. O. Nelson, "A comparison of current measures of the accuracy of feeling-of-knowing predictions," *Psychological Bulletin*, vol. 95, no. 1, pp. 109–133, 1984, doi: 10.1037/0033-2909.95.1.109.

[7] Y. Rosseel, "lavaan: An R package for structural equation modeling," *Journal of Statistical Software*, vol. 48, no. 2, pp. 1–36, 2012, doi: 10.18637/jss.v048.i02.

[8] D. Bates, M. Mächler, B. Bolker, and S. Walker, "Fitting linear mixed-effects models using lme4," *Journal of Statistical Software*, vol. 67, no. 1, pp. 1–48, 2015, doi: 10.18637/jss.v067.i01.

[9] P. Arbuzova *et al.*, "Measuring metacognition of direct and indirect parameters of voluntary movement," *Journal of Experimental Psychology: General*, vol. 150, no. 11, pp. 2208–2229, 2021, doi: 10.1037/xge0000892.

[10] J. W. Bang, M. Shekhar, and D. Rahnev, "Sensory noise increases metacognitive efficiency," *Journal of Experimental Psychology: General*, vol. 148, no. 3, pp. 437–452, 2019, doi: 10.1037/xge0000511.

[11] N. Haddara and D. Rahnev, "The impact of feedback on perceptual decision-making and metacognition: Reduction in bias but no change in sensitivity," *Psychological Science*, vol. 33, no. 2, pp. 259–275, 2022, doi: 10.1177/09567976211032887.

[12] C. Spearman, "Correlation calculated from faulty data," *British Journal of Psychology*, vol. 3, no. 3, pp. 271–295, 1910, doi: 10.1111/j.2044-8295.1910.tb00206.x.

[13] W. Brown, "Some experimental results in the correlation of mental abilities," *British Journal of Psychology*, vol. 3, no. 3, pp. 296–322, 1910, doi: 10.1111/j.2044-8295.1910.tb00207.x.

[14] R Core Team, "R: A Language and Environment for Statistical Computing." Vienna, Austria, 2026.